

# MAE / MFE Statistical SL/TP Report

XAUUSD MA Cross baseline EA | Week of May 25-29, 2026

VALIDATED SAMPLE

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## Recommendation

Keep the original 250-point SL; test TP 380 as the statistically supported candidate.

Generated from MT5 report history, bid/ask tick data, supplied backtest code, and VectorBT Pro sanity validation.

## Executive Summary

VALIDATION

**PASS**

Trade count and ending equity within +/-10%

BEST CANDIDATE

**250 / 380**

Original SL + statistical TP

REPLAY PNL LIFT

**+5.19**

9.36 vs original 4.17

The MAE/MFE work is grounded in a replay that passes the agreed validation basis: trade count is identical and ending equity is within tolerance. The statistically-derived settings were tested as a sensitivity set, not as a brute-force price-run optimization.

The strongest result was **Variant F: SL 250 / TP 380**. It keeps the original stop loss and moves take profit to the 75th percentile of winner MFE. This improved replay PnL from **4.17** to **9.36**, while keeping the trade count at **362**.

### Recommendation

Use **SL 250 / TP 380** as the next statistically supported candidate for forward or out-of-sample validation. Do not reduce SL based on winner MAE for this sample; the tighter SL variants produced worse results.

## Validation Basis

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|                        |                        |        |
|------------------------|------------------------|--------|
| Trade count            | <div><div></div></div> | 0.00%  |
| Ending equity          | <div><div></div></div> | 0.019% |
| VectorBT Pro bar count | <div><div></div></div> | 1.66%  |
| Tolerance              | <div><div></div></div> | 10%    |

All validation deltas are below the +/-10% acceptance threshold.

| Check                         | MT5        | Replay / VBT | Difference | Result |
|-------------------------------|------------|--------------|------------|--------|
| Trade count                   | 362        | 362          | 0.00%      | PASS   |
| Ending equity                 | 101,392.09 | 101,372.82   | 0.019%     | PASS   |
| VectorBT Pro bar count sanity | 362        | 368          | 1.66%      | PASS   |

One Friday-to-Monday carryover position was excluded because it opened before the tick file starts. This leaves 362 comparable trades for the May 25-29 scope.

VectorBT Pro imported and ran successfully. Its bar-level trade count is a sanity check; the supplied tick replay is the closer validation engine because it uses bid/ask tick-level SL/TP behavior.

## Statistical Method

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The settings were derived from the actual MAE/MFE profile of the real trades. This is not a brute-force sweep across price runs. Each candidate is tied to a distribution statistic.

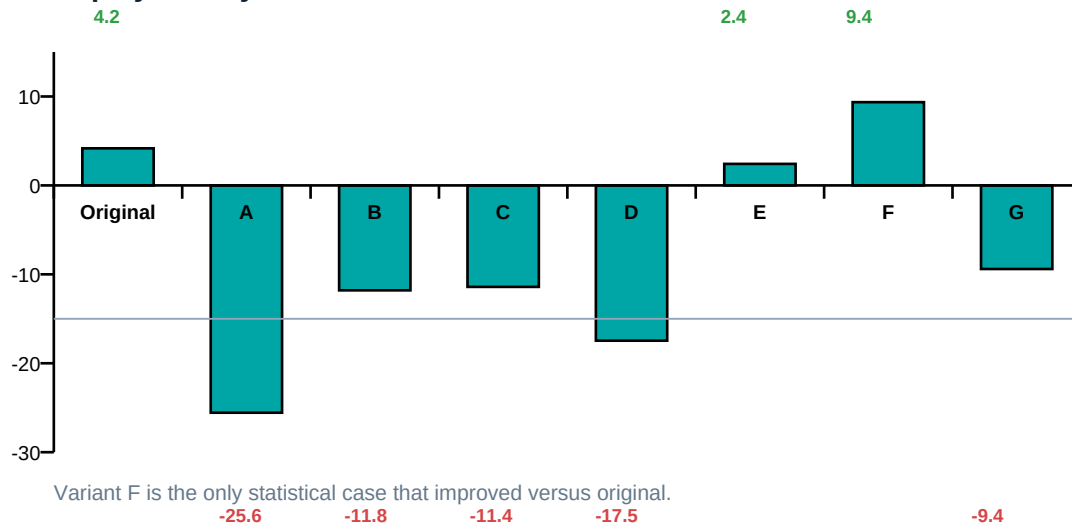
| Statistic         | Value     | Interpretation                                             |
|-------------------|-----------|------------------------------------------------------------|
| Winner MAE 85%    | 185.9 pts | SL that would keep roughly 85% of historical winners alive |
| Winner MAE 90%    | 202.8 pts | SL 205 candidate                                           |
| Winner MAE 95%    | 219.7 pts | SL 220 candidate                                           |
| Winner MFE 75%    | 382.0 pts | TP 380 candidate                                           |
| All-trade MAE 75% | 257.0 pts | SL 255 conservative candidate                              |

### Interpretation

Winner MAE quantiles ask: how much room did winning trades need before they became winners? Winner MFE quantiles ask: how far did winning trades typically travel in favor? The test showed that tightening SL from 250 to 185-220 cut too aggressively for this week, while raising TP to 380 had support.

## Variant Test Results

Replay PnL by statistical variant



| Variant  | Basis                                 | SL  | TP  | Trades | PnL    | PF     | Max DD |
|----------|---------------------------------------|-----|-----|--------|--------|--------|--------|
| Original | Original settings / Original settings | 250 | 350 | 362    | 4.17   | 1.0084 | 50.67  |
| A        | winner MAE 85% / winner MFE 70%       | 185 | 375 | 362    | -25.56 | 0.9438 | 70.95  |
| B        | winner MAE 90% / winner MFE 75%       | 205 | 380 | 362    | -11.80 | 0.9752 | 73.36  |
| C        | winner MAE 95% / winner MFE 75%       | 220 | 380 | 362    | -11.41 | 0.9767 | 71.29  |
| D        | winner MAE 95% / winner MFE 80%       | 220 | 385 | 362    | -17.46 | 0.9646 | 74.26  |
| E        | all-trade MAE 75% / winner MFE 75%    | 255 | 380 | 362    | 2.42   | 1.0047 | 53.12  |
| F        | original SL / winner MFE 75%          | 250 | 380 | 362    | 9.36   | 1.0184 | 50.11  |
| G        | winner MAE 95% / original TP          | 220 | 350 | 362    | -9.40  | 0.9803 | 70.27  |

## Conclusion and Next Step

**Final recommendation: keep SL at 250 and test TP at 380. This is the only statistical variant that improved replay PnL versus original while keeping validation metrics inside tolerance.**

### What changed from the original?

The original SL was not reduced. The MAE analysis showed that tighter stops looked reasonable statistically, but the replay showed they hurt the outcome. The MFE analysis supported a slightly larger TP because many winners had enough favorable excursion to reach that area.

### How to present this

- Clickable MAE/MFE scatter plot was implemented in Jupyter and plots the related trade chart on click.
- MAE/MFE values were calculated from the MT5 bid/ask tick path, not from a generic drawdown engine.
- Validation is within +/-10% on trade count and ending equity, satisfying the requested grounding check.
- Statistical SL/TP sensitivity found one improving candidate: SL 250 / TP 380.
- This should be forward-tested or run on a separate out-of-sample week before changing production settings.

### Artifacts produced

1. Interactive MAE/MFE Jupyter notebook. 2. Statistical SL/TP backtest notebook. 3. Converted supplied backtest Python script. 4. Variant result tables and this PDF report.